MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
FIRST SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MG1MDCSTA100 - Statistical Data Collection

Duration: 1Hour Maximum: 35 Marks

PRACTICAL EXAMINATION

From the following eight questions, students must answer any **five** questions. Each question carries **seven** marks.

Solve the questions using spreadsheet in computer lab . The data in the question along with their answer must be written in the answer paper

1. A real estate agent wants to calculate the average price of houses sold in a neighbourhood to give clients a sense of the typical selling price. Let the house prices in rupees be 600000, 500000, 450000, 380000, 800000, 1200000, 950000, 850000. Help him to do this by calculating arithmetic mean of the data .[Apply][CO7]

2. An IT company wants to examine the salary of the workers to understand overall compensation trends. If 20 workers earn in rupees 16000, 26000, 28000, 9000, 12000, 11500, 24000, 13500, 9500, 10000, 14500, 15000, 12800, 11400, 14700, 18000, 16500, 16400, 30000 and 12000 per month, find the median and mode for the data [Analyse][CO7]

3. A teacher asked the students how many siblings they have and their answers are 2, 1, 3, 2, 3, 4, 1, 3, 5, 4, 3, 3, Find the geometric mean of the data. [Understand][CO7]

4. A shoe company manufactured winter boots with sizes 10, 8, 9, 9, 7, 8, 6, 10, 9, 8, 9, 7, 9. Find harmonic mean of the data.[Apply][CO7]

5. The wickets taken by a bowler in different cricket matches are 3, 5, 2, 1, 4, 5, 2, 4, 1, 2, 2, 3, 5, 2, 3, 3, 2. Find median and harmonic mean of the data. [Analyse][CO7]

6. The information on the observed life times (in hours) of some electrical components are 20, 25, 56, 45, 44, 38, 67, 32, 65, 45, 50, 53, 46, 65, 33, 62, 42, 52. Find the arithmetic mean and harmonic mean of the data. [Understand][CO7]

7. The following data shows the yield of paddy in kilograms from 25 experimental plots: 55, 67.8, 38.5, 62.3, 43.8, 46.3, 55.8, 52.3, 40.5, 38.6, 45.8, 50.7, 44.8, 43.8, 42.1, 54.9, 53.9, 41.8, 65.7, 34.9, 40.6, 47, 45, 54.8, 39.7. Find harmonic mean for the data. [Apply][CO7]

8. The following data shows the number of runs scored by some top batsmen of the world in one day international cricket matches. The scores are 5500, 6700, 3800, 2900, 4700, 8900, 1000, 5600, 5500, 3500, 8800, 8900. Find median and harmonic mean of the data. [Understand][CO7]

9. The number of tourists who visited Ponmudi hills in Trivandrum in a particular month from different states of India are 350, 176, 330, 238, 198, 327, 339, 357, 567, 256, 382, 354, 327, 600, 386,

128, 168, 243, 184, 327, 106, 627, 176. Calculate mode and median for the data. [Apply][CO7]

10. A dietician is analyzing the concentration of a certain nutrient in different food items and recorded value in mg as 1.8, 2.7, 3.6, 4.8, 2.8, 3.9, 2.5, 1.8, 2.8, 3.1, 3.6, 2.5, 1.1, 3.3, 4.2. Find median for the data. [Apply][CO7]

11. A hospital records the waiting times in minutes for patients in getting medicines from pharmacy. Let the waiting times be 6, 9, 10, 4, 3, 7, 2, 8, 10, 12, 2, 5, 7, 8, 3. Find geometric mean and harmonic mean of the data .[Analyse][CO7]

12. The amount of rainfall (in cm) in a particular period at different places are 5.6, 8.7, 3.9, 5, 7.2, 4.9, 5.2, 4.6, 5.3, 5.5, 7.9, 8.3, 9.9, 5.9, 6.9, 7.3 . Find Mean Deviation of the data . [Apply][CO7]

13. A company manufactures screws and packs them . Each pack contains the number of screws as 100, 300, 450, 550, 200, 600, 350, 150, 600, 500, 350, 450, 500, 550, 700. Find the range of the data.[Apply][CO7]

14. The number of smart phones sold at different shops in Cochin on a particular day are 13, 8, 23, 22, 17, 15, 19, 12, 10, 9, 17, 19, 14, 13, 11, 23, 25, 21, 17, 15. Find mean deviation and standard deviation of the data. [Apply][CO7]

15. The temperature recorded in Fahrenheit at different places of Kerala on a particular day were 97, 92, 98, 90, 91, 96, 97, 91, 92, 92, 99, 98. Calculate Standard Deviation and range.[Apply][CO7]

16. The yields in gm from different plants at a village are 10, 14, 8, 16, 15, 17, 20, 22, 18, 16, 19, 16, 10, 20, 22, 32, 31, 26, 25, 24,

21, 27. Find quartile deviation and range of the data.[Understand][CO7]

17. A teacher asked the students to complete 30 pages of a record book. 15 students have completed only 12, 15, 17, 13, 16, 19, 22, 15, 8, 6, 20, 22, 21, 24, 25 pages respectively. Find mean deviation and Standard Deviation.[Analyse][CO7]

18. The marks of students in a class, out of 20, were 14, 15, 14, 18, 17, 15, 20, 11, 10, 16, 15, 16, 18, 19, 12, 20, 20, 13. Find mean deviation.[Apply][CO7]

19. An IT company manager tracks the working hours of employees in a day as 8, 8, 7, 6, 10, 9, 8, 7, 10, 12, 9, 7, 6, 5, 9, 10, 11, 12, 11. Find standard deviation for the data. [Understand][CO7]

20. The time (in seconds) taken by some athletes to complete a 100 metre race are 20, 24, 16, 27, 18, 16, 24, 18, 12, 21, 24, 19, 15, 16, 15, 23, 28, 20, 18, 22, 24, 25. Find Standard Deviation of the data.[Apply][CO7]

21. The number of patients admitted at different hospitals in Kerala who are suffering from TB on a particular day are 22, 14, 16, 18, 20, 22, 17, 18, 19, 17, 11, 19, 17, 15, 15, 17. Find quartile deviation and Standard Deviation of the data.[Analyse][CO7]

22. The daily rainfall in mm at different places are given below. Draw a frequency curve for the data

rainfall	0-3	3-6	6-9	9-12	12-15	15-18	18-21
No. of places	24	20	23	24	18	19	20

[Analyse][CO7]

23. The number of people using each social media platform in a small village are as follows. Draw a pie diagram for this.

Platform	No. of people
instagram	150
facebook	180
whatsapp	250
twitter	90
snapchat	70
telegram	50

[Apply][CO7]

24. The monthly exports of items in a shop is as follows . Draw a simple bar diagram for this

month	Jan	Feb	Mar	Apr	May	June
exports	2500	2000	1500	1300	3000	1400

[Apply][CO7]

25. The following data shows the time (in hours) taken for daily activities by certain professionals in a company. Draw a simple bar diagram based on this data [Analyse][CO7]

activity	work	sleep	leisure	reading	purchase	phone
time	8	6	3	2	2	3

26. The following data shows the market sales of different laptop brands. Draw a pie diagram for this

brand	A	B	C	D	E	F	G
sales	30	45	70	100	25	38	60

[Apply][CO7]

27. The following data gives the distribution of marks of students in a class. Draw a frequency polygon for this

marks	0-8	8-16	16-24	24-32	32-40
students	20	8	8	12	15

[Analyse][CO7]

28. The following data shows the commute times (in minutes) of some professionals in a company. Draw ogives for this data

time	0-5	5-10	10-15	15-20	20-25	25-30	30-35
freq	7	12	15	20	5	3	2

[Apply][CO7]

29. The following data shows the times (in minutes) taken for online education by some students in a college in a day. Draw a less than ogive for this data

time	0-10	10-20	20-30	30-40	40-50	50-60	60-70
freq	5	11	15	16	9	7	4

[Apply][CO7]

30. The data about the website visits by people of different age groups are given. Draw a greater than ogive for the data

age	10-15	15-20	20-25	25-30	30-35	35-40	40-45
No. of people	5	12	16	20	14	9	2

[Apply][CO7]

31. The following data shows the number books read by different persons in a year, from a library. Draw ogives for the data

books	0-4	4-8	8-12	12-16	16-20	20-24	24-28	28-32
No. of people	10	22	30	40	25	17	8	7

[Apply][CO7]

32. The following data shows the sleeping hours of some doctors in a hospital. Draw a greater than ogive for this.

hours	2-4	4-6	6-8	8-10	10-12	12-14
freq	14	26	38	45	33	22

[Apply][CO7]

33. The following data shows the hours spent by students in a school to do their home works in a week. Draw a frequency curve for this data

hours	0-3	3-6	6-9	9-12	12-15	15-18
freq	12	20	25	15	8	4

[Analyse][CO7]

34. The following data shows the weight of packages (in pounds) shipped from an international company to different parts of world. Draw a frequency curve for this data

weight	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No of packages	16	30	40	25	18	16	3

[Apply][CO7]

35. The following data shows the types of pets owned by different families in a city. Draw a pie diagram for this

Pet type	No. of families
cats	45
Love birds	50
fish	30
parrot	20
others	10

[Apply][CO7] (5 x 7 = 35)